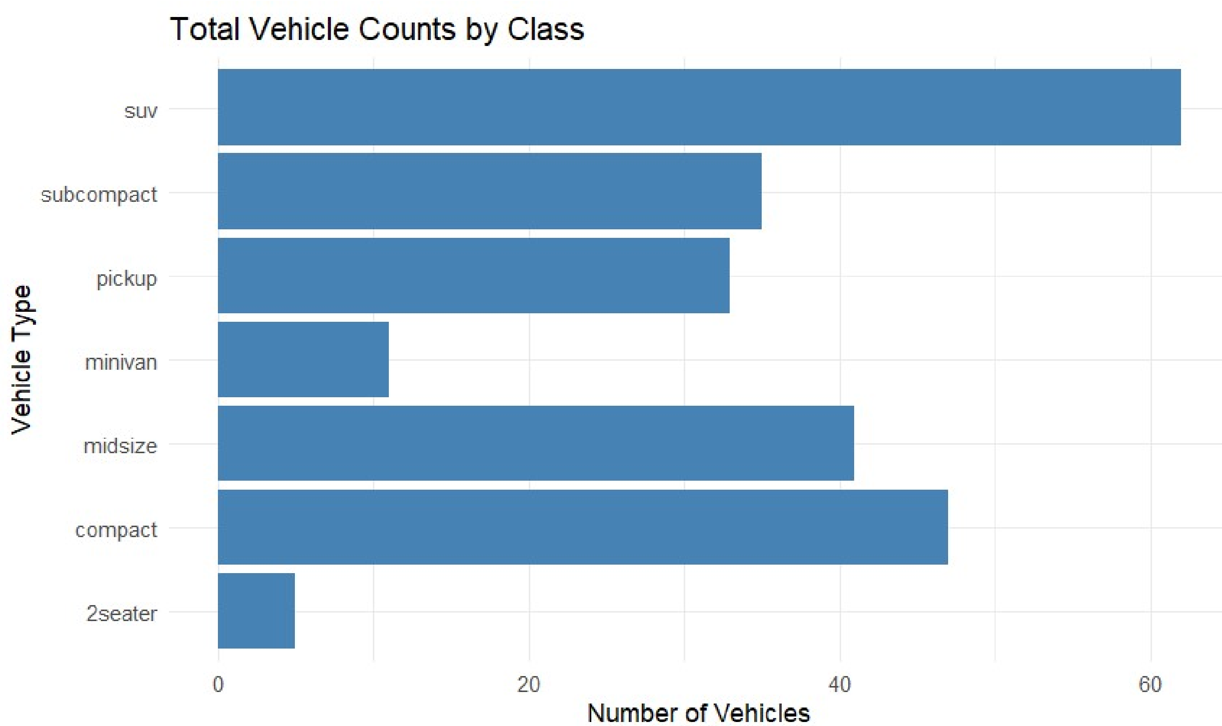
Yiscah Mark

This is the code for the bar chart comparing the amount per vehicle type.

```
ggplot(mpg, aes(y = class)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Total Vehicle Counts by Class",
       x = "Number of Vehicles",
       y = "Vehicle Type") +
  theme_minimal()
```



The following is code for a bar chart that is grouped by 4-wheel drive, front, and rear. Each bar is colored to represent the number of each wheel type.

```
ggplot(mpg, aes(x = class, fill = drv)) +
  geom_bar() +
  labs(title = "Drive Train Distribution by Class",
       x = "Vehicle Class",
       y = "Count",
       fill = "Drive Train") +
```

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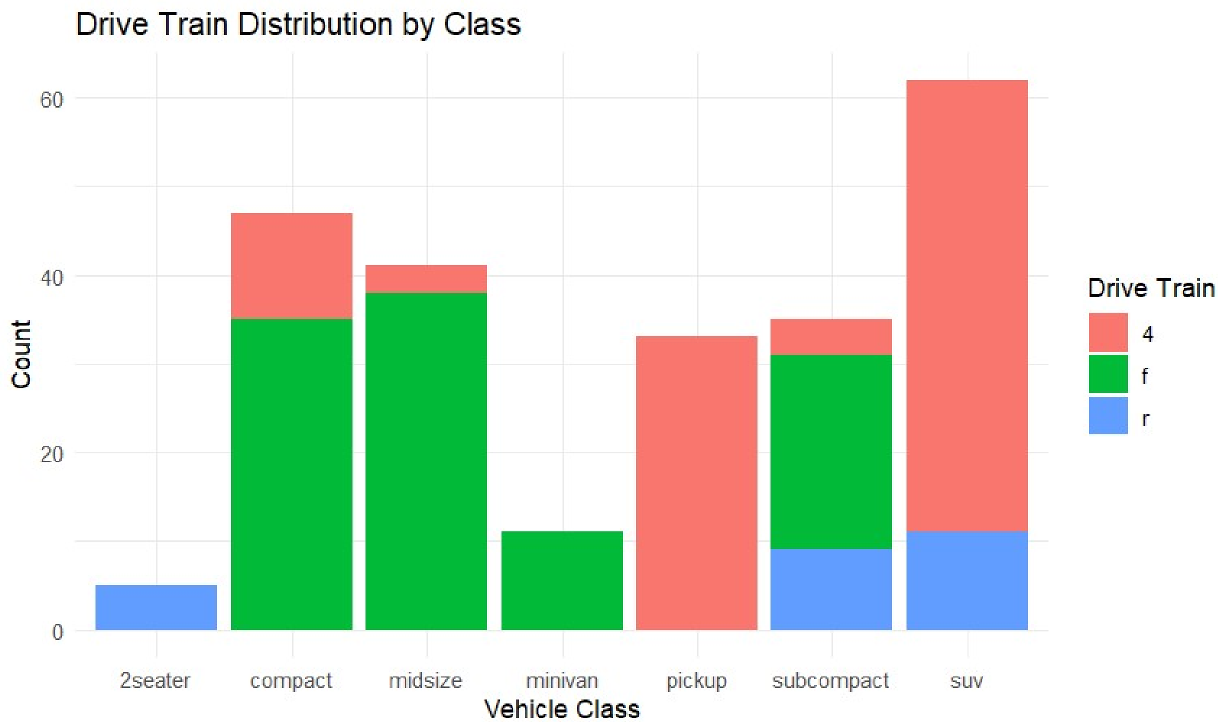
```
  theme_minimal()
```



This is the code for the heatplot. The heat plot has a variable for the number of cylinders in every car type, as well as a third variable – the miles per gallon, which is shown with color. There is a scale near it to show what each color represents.

```
ggplot(mpg, aes(x = factor(cyl), y = class, fill = cty)) +
  geom_tile() +
  scale_fill_viridis_c() +
  labs(title = "Heat Map: Average City MPG",
      x = "Cylinders",
      y = "Vehicle Class",
      fill = "City MPG") +
  theme_minimal
```

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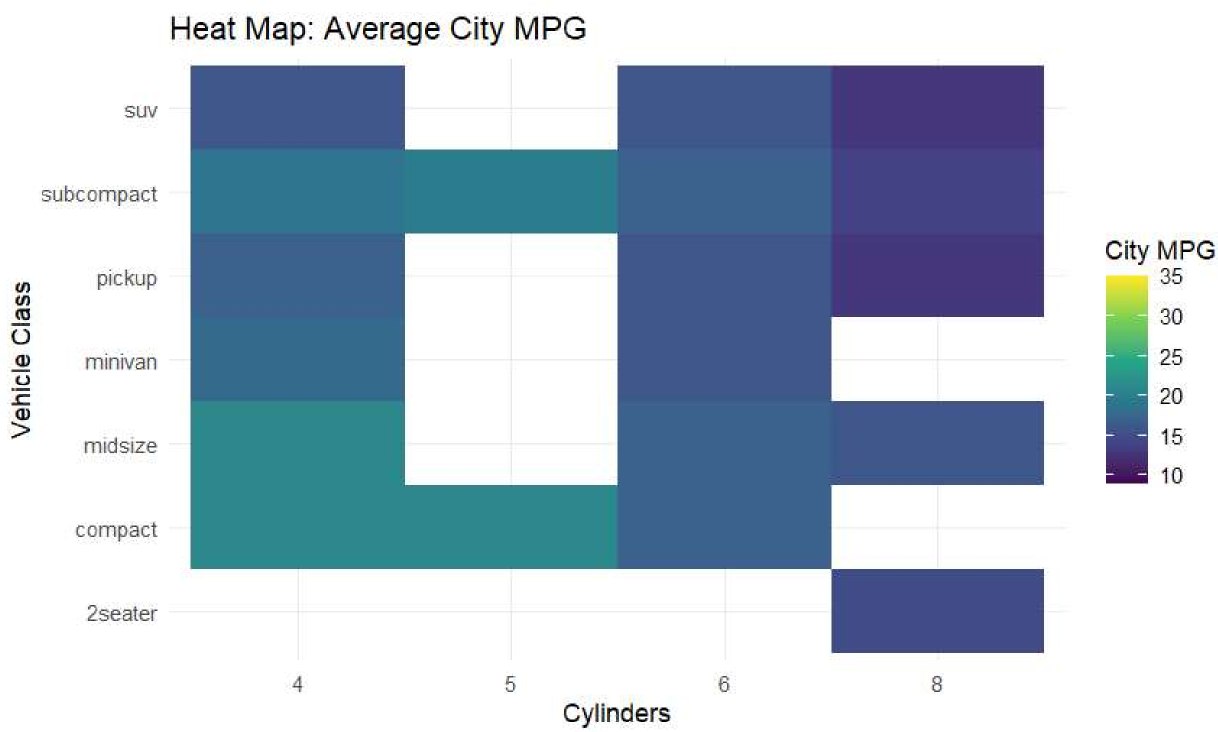
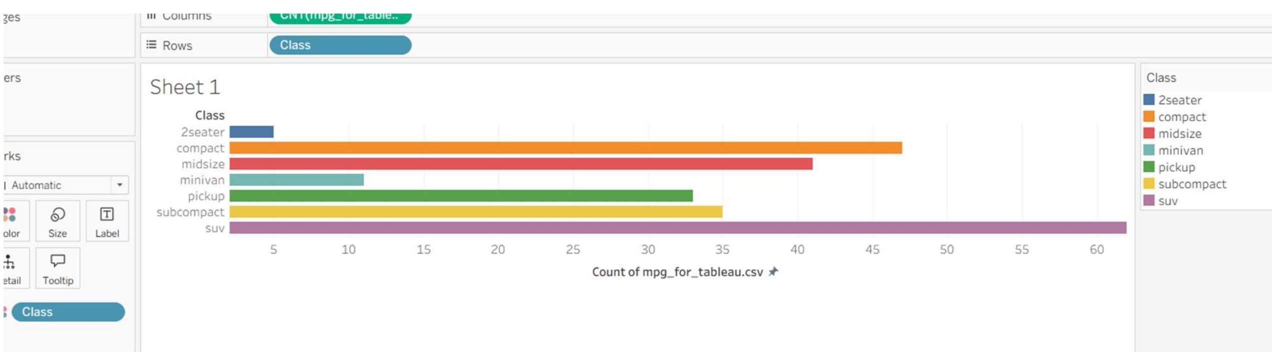


The following are 3 diagrams from Tableau: a bar chart, a grouped bar chart, and a heat chart, respectively. With Tableau, instead of code, categories are just dropped into rows or columns, and specific adjustments are made. In the first chart, the color has no real meaning. In the second step, it groups the cars in each type by wheel type.

At first, the heat map that I created in Tableau appeared very different from the one in R. I had to change the dimensions, as well as the size of the squares, and now I see that it is the same.



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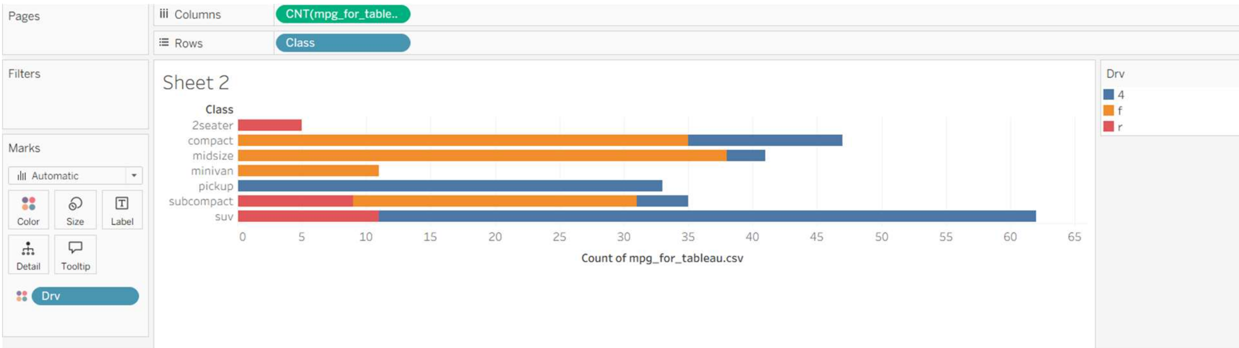
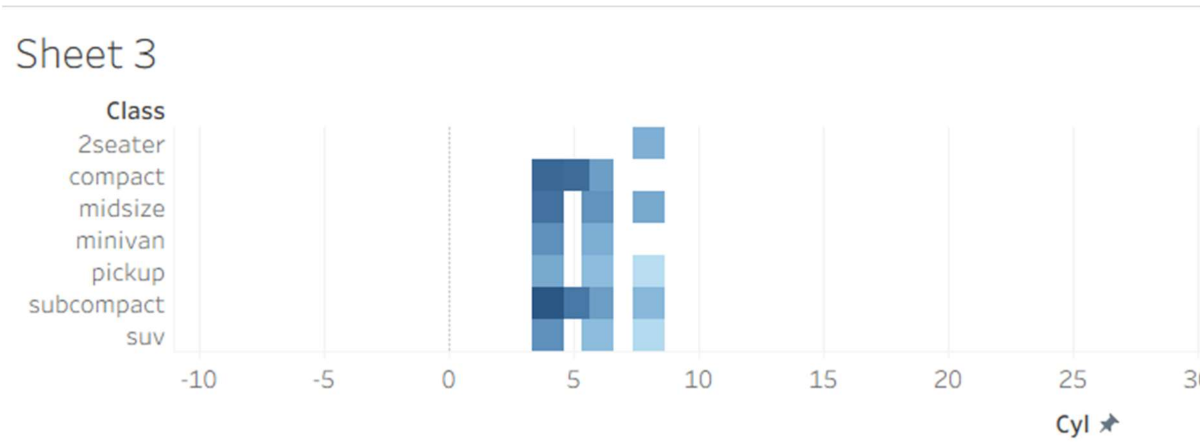


Pages
Columns
CNT(mpg_for_table..
Rows
Class
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Drv
Sheet 2
Class
2seater
compact
midsize
minivan
pickup
subcompact
suv
0 5 10 15 20 25 30 35 40 45 50 55 60 65
Count of mpg_for_tableau.csv
Drv
4
f
r



Sheet 3
Class
2seater
compact
midsize
minivan
pickup
subcompact
suv
-10 -5 0 5 10 15 20 25
Cyl